2.4 Probability Distributions

Mathematics B (MEI) — OCR B — A-Level (H640) — Statistics — spec refs R1–R13

What this topic covers. Discrete random variables and probability functions, the binomial distribution as a model, the Normal distribution including standardising and the continuity correction, and judging when each model is appropriate.

1. Discrete random variables

A **discrete random variable** X takes separate values with given probabilities. Capital X denotes the variable; lower-case x or r denotes a particular value (spec ref R6).

A probability function may be given algebraically or in a table. Every valid distribution satisfies

$$\sum \mathrm{P}(X = x) = 1$$

Worked example. $\mathrm{P}(X = x) = kx$ for $x = 1, 2, 3, 4$. Find k.

$$k(1 + 2 + 3 + 4) = 10k = 1 \implies k = 0.1$$

A **discrete uniform** distribution is one in which all outcomes are equally likely — the score on a fair die, for instance.

2. The binomial distribution

$X \sim \mathrm{B}(n, p)$ counts the number of successes in n trials. The symbol $\sim$ is read "has the distribution".

When a binomial model applies

The specification lists the conditions explicitly (R1):

- The trial is conducted a **fixed** number of times.
- There are exactly **two** outcomes, "success" and "failure".
- The probability of success is the **same** each time.
- The trials are **independent** of one another.

Writing $q = 1 - p$, the probability of exactly r successes is

$$\mathrm{P}(X = r) = \binom{n}{r} p^r q^{n-r}$$

Worked example. $X \sim \mathrm{B}(10, 0.4)$. Find $\mathrm{P}(X = 3)$.

$$\binom{10}{3}(0.4)^3(0.6)^7 = 120 \times 0.064 \times 0.0279936 \approx 0.215$$

Mean and expected frequencies

$$\text{mean} = np$$

For $X \sim \mathrm{B}(20, 0.3)$ the mean is $20 \times 0.3 = 6$. In 50 trials with $p = 0.2$, the expected number of successes is 10.

Use the complement for "at least one": $\mathrm{P}(X \geq 1) = 1 - \mathrm{P}(X = 0) = 1 - q^n$. For $X \sim \mathrm{B}(5, 0.2)$ this is $1 - 0.8^5$.

A binomial model **fails** when sampling without replacement from a small population, because the probability of success changes between trials. It remains a reasonable *approximation* when the population is very large relative to the sample.

3. The Normal distribution

$X \sim \mathrm{N}(\mu, \sigma^2)$ describes a continuous variable whose distribution is symmetric and bell-shaped, with mean μ and standard deviation σ.

Note that the second parameter is the **variance**. For $\mathrm{N}(100, 25)$ the standard deviation is 5, not 25.

Properties of the curve

- Symmetric about μ, where the mean, median and mode coincide.
- The total area under the curve is 1, and area represents **probability**.
- $\mathrm{P}(X > \mu) = 0.5$.
- The **points of inflection** lie at $\mu \pm \sigma$, one standard deviation either side of the mean (R11).
- Roughly 68% of values lie within one standard deviation of the mean, and about 95% within two.

Standardising

$$Z = \frac{X - \mu}{\sigma}, \qquad Z \sim \mathrm{N}(0, 1)$$

Worked example. $X \sim \mathrm{N}(60, 16)$. Find the z-value for $X = 68$.

Here $\sigma = \sqrt{16} = 4$, so $z = \dfrac{68 - 60}{4} = 2$.

Worked example. $X \sim \mathrm{N}(80, 25)$. Find X when $z = -1.6$.

$$X = 80 + (-1.6)(5) = 72$$

Linear transformations

A linear transformation of a Normal variable is again Normal (R10). If $Y = aX + b$ then

$$\text{mean of } Y = a\mu + b, \qquad \text{variance of } Y = a^2\sigma^2$$

Worked example. $X \sim \mathrm{N}(50, 16)$ and $Y = 3X + 2$.

$$\text{mean} = 3(50) + 2 = 152, \qquad \text{variance} = 3^2 \times 16 = 144$$

So $Y \sim \mathrm{N}(152, 144)$.

The multiplier is **squared** in the variance but not in the mean, and the additive constant b affects the mean only — shifting a distribution does not change its spread.

4. The continuity correction

When a **continuous** Normal distribution is used to model **discrete** data, each integer is treated as covering the interval half a unit either side. This adjustment is the **continuity correction** (R8).

Discrete statement	Normal equivalent
$\mathrm{P}(X \le 10)$	$\mathrm{P}(X < 10.5)$
$\mathrm{P}(X < 10)$	$\mathrm{P}(X < 9.5)$
$\mathrm{P}(X \ge 12)$	$\mathrm{P}(X > 11.5)$
$\mathrm{P}(X > 12)$	$\mathrm{P}(X > 12.5)$
$\mathrm{P}(X = 12)$	$\mathrm{P}(11.5 < X < 12.5)$

Decide the direction by asking which integers the statement should **include**. $\mathrm{P}(X \ge 12)$ must include 12, so the boundary goes below it, at 11.5. Sketching the discrete bars against the curve settles it every time.

MEI expects you to *recognise from the shape* of a binomial distribution when a Normal approximation is reasonable — a roughly symmetric, bell-shaped histogram. Learning a formal list of conditions for the approximation is not required by this specification.

5. Choosing and criticising a model

Modelling with probability includes recognising when the binomial or Normal model is **not** appropriate (R13).

Situation	Model	Why
Number of heads in 20 tosses	Binomial	Fixed trials, two outcomes, constant p, independent
Number of sixes in 30 rolls	$\mathrm{B}(30, \frac{1}{6})$	Same conditions hold
Heights of adults	Normal	Continuous, symmetric, clustered about a mean
Drawing from a small bag without replacement	Neither cleanly	p changes between trials
Markedly skewed data	Not Normal	The Normal curve is symmetric

Criticising a model means examining whether its **assumptions** hold in the given context — not recalculating. Say which condition is doubtful and why it matters.

6. Common pitfalls

- Reading the second Normal parameter as the standard deviation instead of the variance.
- Forgetting to square the multiplier when transforming a variance.
- Applying the continuity correction in the wrong direction.
- Using a binomial model where the probability of success changes between trials.
- Forgetting the $\binom{n}{r}$ coefficient in the binomial formula.
- Computing $\mathrm{P}(X \geq 1)$ term by term instead of using $1 - q^n$.
- Placing the points of inflection anywhere other than $\mu \pm \sigma$.